16 Comparative Music Cognition: Cross-Species and Cross-Cultural Studies

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I. Introduction

Music, according to the old saw, is the universal language. Yet a few observations quickly show that this is untrue. Our familiar animal companions, such as dogs and cats, typically show little interest in our music, even though they have been domesticated for thousands of years and are often raised in households where music is frequently heard. More formally, a scientific study of nonhuman primates (tamarins and marmosets) showed that when given the choice of listening to human music or silence, the animals chose silence (McDermott & Hauser, 2007). Such observations clearly challenge the view that our sense of music simply reflects the auditory system's basic response to certain frequency ratios and temporal patterns, combined with basic psychological mechanisms such as the ability to track the probabilities of different events in a sound sequence. Were this the case, we would expect many species to show an affinity for music, since basic pitch, timing, and auditory sequencing abilities are likely to be similar in humans and many other animals (Rauschecker & Scott, 2009). Hence although these types of processing are doubtlessly relevant to our musicality, they are clearly not the whole story. Our sense of music reflects the operation of a rich and multifaceted cognitive system, with many processing capacities working in concert. Some of these capacities are likely to be uniquely human, whereas others are likely to be shared with nonhuman animals. If this is true, then no other species will process music *as a whole* in the same way that we do. Yet certain aspects of music cognition may be present in other species, and this is important for music psychology. As we shall see in this chapter, a systematic exploration of the commonalities and differences between human and nonhuman music processing can help us study the evolutionary history of our own musical abilities.

Turning from other species to our own, is the "music as universal language" idea any more valid? The answer is still no, though the evidence is more mixed.

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For example, it is easy to find Westerners, even highly trained musicians, who have little response (or even an aversive response) to music that is greatly valued in other cultures. They might recognize it as music and even formulate some sense of its meaning, but such formulations often rely on more general surface qualities of the music without an awareness of deeper structures. Of course, there is a great deal of boundary-crossing and blending in music around the world, especially in popular and dance music, and there are certain basic musical forms, such as lullabies, which show a good deal of cross-cultural similarity (Unyk, Trehub, Trainor & Schellenberg. 1992). Nevertheless, it is clear that blanket statements about music as a universal language do not hold, and this is true when dealing with "folk" music, as well as "art" music. (NOTE: As a simple and informal test of this premise, visit the Smithsonian Folkways website and listen to folk music clips from 20 or 30 cultures around the world). This points to an enormously important feature of human music: its great diversity. Music psychology has, until recently, largely ignored this diversity and focused almost entirely on Western music. This was a natural tendency given that most of the researchers in the field were encultured to Western musical styles. Unfortunately, theories and research findings based solely on a single culture's music are severely limited in their ability to tell us about music cognition as a global human attribute. This is why comparative approaches to music psychology, although relatively new, are critical to our understanding of music cognition.

II. Cross-Species Studies

A. *Introduction*

Cross-species research on music cognition is poised to play an increasingly important role in music psychology in the 21st century. This is because such studies provide an empirical approach to questions about the evolutionary history of human music (Fitch, 2006; McDermott & Hauser, 2005). Music cognition involves many distinct capacities, ranging from "low-level" capacities not specific to music, such as the ability to perceive the pitch of a complex harmonic sound, to "high-level" capacities that appear unique to music, such as the processing of tonal-harmonic relations on the basis of learned structural norms (Koelsch, 2011; Peretz & Coltheart, 2003). It is very unlikely that all of these capacities arose at the same time in evolution. Instead, the different capacities are likely to have different evolutionary histories. Cross-species studies can help illuminate these histories, using the methods of comparative evolutionary biology (see Fitch, 2010, for an example of this approach applied to the evolution of language). For example, the ability to perceive the pitch of a complex harmonic sound, a basic aspect of auditory perception, is likely to be a very ancient ability. Comparative studies suggest that this ability is widespread among mammals and birds, and is present in a variety of fish species (Plack, Oxenham, Fay, & Popper, 2005). This suggests that basic pitch perception has a long evolutionary history, far predating the origin of humans.

Furthermore, it means that we can study commonalities in how living animals use this ability in order to glean ideas about *why* the ability evolved. For example, if many species use pitch for recognizing acoustic signals from other organisms and for identifying and tracking individual objects in an auditory scene (Bregman, 1990; Fay, 2009), then these functions may have driven the evolution of basic pitch perception.

On the other hand, consider the ability to perceive abstract structural properties of tones, such as the sense of tension or repose that enculturated listeners' experience when hearing pitches in the context of a musical key (e.g., the perceived stability of a pitch, say A440, when it functions as the tonic in one key, vs. the perceived instability of this same pitch when it functions as the leading tone in a different key, cf. Bigand, 1993). This ability seems music-specific (Peretz, 1993), and we have no idea if nonhuman animals (henceforth "animals") experience these percepts when they hear human music. It is possible that such percepts reflect implicit knowledge of tonal hierarchies, that is, hierarchies of pitch stability centered around a tonic or most stable note (Krumhansl, 1990). According to one current theory (Krumhansl & Cuddy, 2010), two basic processing mechanisms underlie the formation of tonal hierarchies: the use of cognitive reference points and statistical learning based on passive exposure to music. There is no a priori reason to suspect that the use of cognitive reference points and statistical learning are unique to humans, as these are very general psychological processes. Imagine, however, that comparative research shows that animals raised with exposure to human music do *not* develop sensitivity to the abstract structural qualities of musical tones. We could then infer that this aspect of music cognition reflects special features of human brain function, on the basis of brain changes that occurred since our lineage diverged from other apes several million years ago. The hunt is then on to determine what unique aspects of human brain processing support this ability, and why we have this ability.

In the preceding hypothetical examples, an aspect of music cognition was either widespread across species or uniquely human, and each of these outcomes had implications for evolutionary issues. There is, however, another possible outcome of comparative work: an aspect of music cognition can be shared by humans and a select number of other species. For example, Fitch (2006) has noted that drumming is observed in humans and African great apes (such as chimpanzees, which drum with their hand on tree buttresses), but not in other apes (such as orangutans) or non-ape primates. If this is the case, then it suggests that the origins of drumming behavior in our lineage can be traced back to the common ancestor of humans and African great apes. This sort of trait sharing, due to descent from a common ancestor with the trait, is known as "homology" in evolutionary biology. Another type of sharing, based on the independent evolution of a similar trait in distantly related animals, is called "convergence." A recent example of convergence in music cognition is the finding that parrots spontaneously synchronize their movements to the beat of human music (Patel, Iversen, Bregman, & Schulz, 2009), even though familiar domestic animals such as dogs and cats (who are much more closely related to humans) show no sign of this behavior. Cases of convergence provide important grounds for formulating hypotheses about why an aspect of music cognition arose in our species. If a trait appears in humans and other distantly related species, what do

humans and those species have in common that could have led to the evolution of the trait? For example, it has been proposed that the capacity to move to a musical beat arose as a fortuitous byproduct of the brain circuitry for complex vocal learning, a rare ability that is present in humans, parrots, and a few other groups, but absent in other primates. Complex vocal learning is associated with special auditory-motor connections in the brain (Jarvis, 2007), which may provide the neural foundations for movement to a beat (Patel, 2006). This hypothesis suggests that movement to a musical beat may date back to the origins of vocal learning in our lineage (i.e., possibly before *Homo sapiens*, cf. Fitch 2010). Furthermore, the hypothesis makes testable predictions, such as the prediction that vocal nonlearners (e.g., dogs, cats, horses, and chimps) cannot be trained to move in synchrony with a musical beat, because they lack the requisite brain circuitry for this ability.

We have discussed three possible outcomes of cross-species studies of music cognition: a component of music cognition can be (1) widespread across species, (2) restricted to humans and some other species, or (3) uniquely human. These three categories provide a framework for classifying cross-species studies of music cognition. The goal of this part of the chapter is to discuss some key conceptual issues that arise when a component of music cognition is placed in one of these three categories. That is, the goal is to bring forth issues important for future research, rather than to provide an exhaustive review of past research. Hence each of the categories is illustrated with a discussion of a few selected studies. These studies were chosen because they raise questions that can be studied immediately, using available methods for research on animals.

B. Abilities That are Widespread among Other Species

When an ability is widespread among species, one can conclude that it is very ancient (see the example of basic pitch perception at the start of the chapter). For example, Hagmann and Cook (2010) recently showed that pigeons could easily discriminate two isochronous tone sequences on the basis of differences in tempo and could generalize this discrimination to novel tempi. Similarly, McDermott and Hauser (2007) showed that monkeys (tamarins and marmosets) discriminated between slow and fast click trains. Indeed, it seems likely that basic auditory tempo discrimination is widespread among vertebrates, given that differences in sound rate are important for identifying a variety of biological and environmental sounds. This in turn implies that this ability is (1) not specific to music and (2) was present early in vertebrate evolution. In other words, music cognition built on this preexisting ability.

Of course, human music cognition may have elaborated on this ability in numerous ways. For example, the human sense of tempo in music typically comes from a combination of the rate of a perceived beat (extracted from a complex musical texture based on patterns of accent and timing) and the rate of individual events at the musical surface (London, 2004). Hence the demonstration of basic tempo discrimination in another animal based on isochronous tones or clicks does not necessarily mean that the animal could discriminate tempo in human music, or that the animal would perceive the same tempo as a human listener when listening to music. This leads to the

first conceptual point of this section: even when an ability is widespread, it may have been refined in human evolution in a way that distinguishes us from other animals.

To further illustrate this point, consider basic pitch processing. When humans process a complex periodic sound consisting of integer harmonics of a fundamental frequency (such as a vowel or cello sound), they perceive a pitch at the fundamental frequency, even if that frequency is physically absent (the "missing fundamental"). Hence the nervous system *constructs* the percept of pitch from analysis of a complex physical stimulus (Cariani & Delgutte, 1996; McDermott & Oxenham, 2008). This ability is likely to be widespread among mammals and birds: monkeys, birds, and cats have all been shown to perceive the missing fundamental, and recent electrophysiological work has revealed "pitch-sensitive" neurons in the monkey brain, in a region adjacent to primary auditory cortex (Bendor & Wang, 2006).

However, a salient feature of missing fundamental processing in humans is that it shows a right-hemisphere bias (Patel & Balaban, 2001; Zatorre 1988). Zatorre, Belin, and Penhune (2002) have suggested that the right-hemisphere bias in human pitch processing reflects a tradeoff in specialization between the right and left auditory cortex (rooted in neuroanatomy), with right-hemisphere circuits having enhanced spectral resolution and left-hemisphere circuits having enhanced temporal resolution (cf. Poeppel, 2003). If this is correct, then was this tradeoff driven by the rise of linguistic and musical communication in our species? Or is the asymmetry widespread in other mammals and birds, suggesting that it existed before human language and music? At present, we do not know if there is a hemispheric asymmetry for missing fundamental processing in other animals, but the question is amenable to empirical research.

A second conceptual point about widespread abilities concerns the use of species-appropriate stimuli in music-cognition research. Cross-species studies of music cognition typically employ human music, but this may not always be the best approach, depending on the hypothesis one is testing. For example, Snowdon and Teie (2010) conducted a study with tamarin monkeys to test the hypothesis that one source of music's emotional power is the resemblance of musical sounds to affective vocalizations. To test this hypothesis in a species-appropriate way, the researchers created novel pieces for cello based on the pitch and temporal structure of tamarin threat or affiliative vocalizations, and then played these to tamarins in the laboratory. The researchers found that tamarins showed increased arousal to threat-based music, and increased calm behavior to the affiliation-based music. This suggests that tamarins were reacting to abstract versions of their own, species-specific emotional sounds, presented via a musical instrument. This sort of study could be extended to other species (e.g., dogs, cats), using their own emotional vocalizations as a source of compositional material. An interesting question for such research is whether musicalized versions of the vocalizations are ever *more* potent than actual vocalizations in terms of eliciting emotional responses, that is, if they can act as a "superstimulus" by isolating key acoustic features of emotional vocalizations and exaggerating them, as has been suggested for human musical instruments (Juslin & Laukka, 2003). In examining emotional responses to music

in animals, future work will benefit from measuring physiological variables. For example, the stress hormone cortisol and the neuropeptide oxytocin could be measured, since these have been shown to be modulated by soothing music in randomized controlled studies of humans (Koelsch et al., 2011; Bernatzky, Presh, Anderson, & Panksepp, 2011).

C. Abilities Restricted to Humans and Select Other Species

Some components of music cognition may exist in humans and a few select other species. For example, 6-month-old human infants prefer consonant to dissonant musical sounds (Trainor & Heinmiller, 1998) (although this finding is from Western-enculturated infants and needs to be tested in other cultures). In contrast, tamarin monkeys show no such preferences when tested in an apparatus designed for the study of animal responses to music (McDermott & Hauser, 2004) (Figure 1).

However, a 5-month old human-raised chimpanzee did show a preference for consonant over dissonant music (Sugimoto et al., 2010), as did newly hatched domestic chicks (Chiandetti & Vallortigara, 2011). Interestingly, both of these

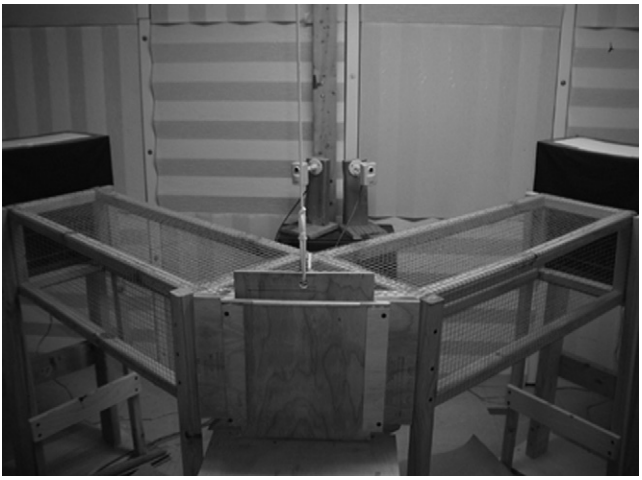

Figure 1 An apparatus used to test musical preferences in a nonhuman primate. The apparatus consists of a V-shaped maze elevated a few feet off the floor. The maze has two arms, which meet at a central point at which the animal is released into the maze. An audio speaker is located at the end of each branch of the maze. After the animal is released into the entrance of the maze, the experimenter leaves the room and raises the door to the maze via a pulley. Whenever the animal enters one arm of the maze, the experimenter begins playback of sounds from the speaker on that arm. The two speakers produce different sounds (e.g., consonant vs. dissonant chord sequences), and the animal thus controls what sounds it hears by its position in the maze (no food rewards are given). Testing continues for some fixed length of time (e.g., 5 minutes) and is videotaped for later analysis. The amount of time spent in each arm is taken as a measure of preference for one sound over the other. From McDermott and Hauser (2004), reproduced with permission. ©2004 Elsevier.

latter studies used juvenile animals with no prior exposure to music, raising the question of whether there is a widespread initial bias for consonant sounds in young mammals and birds. Restricting the discussion to primates, however, the contrast between the findings with monkeys (tamarins) and apes (chimpanzees) is intriguing. If this distinction is maintained in future research, it would suggest that a preference for consonant musical sounds is restricted to great apes among primates. (Further research with other primate species is needed to test such an idea. Among monkeys, marmosets would be a good choice because they have a complex acoustic communication with various "tonal" calls (cf. Miller, Mandel, & Wang, 2010.) If further research supports an ape-specific preference for consonant musical sounds among primates, this would raise interesting questions about why such a predisposition evolved in the ape lineage. (As a methodological note, however, it remains unclear to what extent the preference observed in human infant studies is due to prior exposure to Western music, since the fetus can hear in utero and can learn musical patterns before birth, cf. Patel, 2008, pp. 377−387.)

As with the example of ape drumming mentioned earlier, if a component of music cognition is found only in humans and other apes (but not in non-ape primates), this suggests the component is inherited from the common ancestor of humans and apes. Of course, this does not necessarily mean that this ancestor used this component as part of music-making. Drumming, for example, may have originally had a nonmusical function, which was later modified by members of our own lineage for musical ends, after our lineage split from other apes. This leads to our first conceptual point for this section: when a component of music cognition is shared by homology with other apes, we cannot conclude that the common ancestor was making music. However, we can look for common patterns in how living apes use this ability to get ideas about the original function of this component in ape evolution. For example, chimps and gorillas use manual drumming as part of acoustic-visual displays indicating dominance, aggression, or an invitation to play (Fitch, 2006), and this may hold clues to the original function of ape drumming (cf. Merker, 2000). Similarly, an ape-specific preference for consonant musical sounds may have its roots in a predisposition for attending to (nonmusical) harmonic vs. inharmonic sounds. McDermott, Lehr, and Oxenham (2010) recently showed that a preference for consonant over dissonant musical intervals in humans is correlated with a preference for harmonic spectra (i.e., spectra with integer-ratio relations between frequency components). If ape vocalizations (and other naturally occurring resonant sources) are rich in such sounds, this could explain the evolution of a perceptual bias toward such sounds.

In contrast to examples of trait-sharing based on inheritance from a common ancestor, humans can also share components of music cognition with distantly related species, that is, via convergent evolution (cf. Tierney, Russo, & Patel, 2011). As noted in the introduction, humans and parrots share an ability to synchronize their movements to a musical beat, even though animals more closely related to humans, such as dogs, cats, and other primates, do not seem to have this ability (Patel et al., 2009; Schachner, Brady, Pepperberg, & Hauser, 2009). It should be noted, however, that controlled experiments attempting to teach dogs, cats, and

primates to move to a musical beat remain to be done. (Indeed, there is only one scientific study in which researchers have tried to train nonhuman mammals to move in synchrony with a metronome. Notably, the animals [rhesus monkeys] were unsuccessful at this task despite more than a year of intensive training [Zarco, Merchant, Prado, & Mendez, 2009]. This stands in contrast to a recent laboratory study with small parrots [budgerigars], who learned to entrain their movements to a metronome at several different tempi [Hasegawa, Okanoya, Hasegawa, & Seki 2011].)

Why would humans and parrots share the ability to synchronize to a musical beat? This behavior involves a tight coupling between the auditory and motor systems of the brain, since the brain must anticipate the timing of periodic beats and communicate this information dynamically to the motor system, in order for synchronization to occur. It is known that complex vocal learning, which exists in humans, parrots, and a few other groups, but not in other primates, leads to special auditory-motor connections in the brain (Jarvis, 2007). (Complex vocal learning is the ability to mimic complex, learned sounds with great fidelity). According to the "vocal learning and rhythmic synchronization hypothesis" (Patel, 2006), the auditory-motor connections forged by the evolution of vocal learning also support movement to a musical beat. Importantly, current comparative neuroanatomical research points to certain basic similarities in the brain areas and connections involved in complex vocal learning in humans and birds (Jarvis, 2007, 2009). That is, despite the fact that complex vocal learning evolved independently in humans, parrots, and some other groups (e.g., dolphins, songbirds), there may be certain developmental constraints on vertebrate brains such that vocal learning always evolves using similar brain circuits. If this is the case, then vocal learning in birds and humans may be a case of "deep homology," that is, a trait that evolved independently in distant lineages yet is based on similar underlying genetic and neural mechanisms (Shubin, Tabin, & Carroll, 2009).

This leads to the second conceptual point of this section: when a nonhuman animal shares a behavioral ability with humans, it is important to ask if this is based on similar underlying neural circuits to humans, or if the animal is producing the ability by using very different neural circuits. This question is particularly important when dealing with species that are distantly related to humans (such as birds). If the animal is using quite different neural circuits, then this limits what we can infer about the factors that led to the evolution of this trait in humans. For example, some parrots can "talk" (emulate human speech). Yet when parrots produce words, there is little doubt that the underlying brain circuitry has many important differences from human linguistic processing, because humans integrate rich semantic and syntactic processing with complex vocal motor control.

D. Abilities That Are Uniquely Human

Components of music cognition that are uniquely human are among the most interesting from the standpoint of debates over the evolution of human music. Do they reflect the existence of brain networks that have been specialized over evolutionary

time for musical processing? Or did these components arise in the context of other cognitive domains and then get "exapted" (or "culturally recycled") by humans for musical ends (Dehaene & Cohen, 2007; Gould & Vrba, 1982; Justus & Hutsler, 2005; Patel, 2010)?

To take one example, humans show great facility at recognizing melodies that have been shifted up or down in frequency. For example, we can easily recognize the "Happy Birthday" tune whether played on a piccolo or a tuba. This is because humans rely heavily on *relative pitch* in tone sequence recognition (Lee, Janata, Frost, Hanke, & Granger, 2011). A reliance on relative pitch is a basic component of music perception, and surprisingly, may be uniquely human (McDermott & Oxenham, 2008). Extensive research with songbirds has shown that they have great difficulty recognizing tone sequences that have been shifted up or down in frequency, even with extensive training. It appears that unlike most humans, songbirds gravitate toward absolute pitch cues in recognizing tones or tone sequences, and make very limited use of relative pitch cues (Page, Hulse, & Cynx, 1989; Weisman, Njegovan, Williams, Cohen, & Sturdy, 2004), a fact that surprised birdsong researchers (Hulse & Page, 1988). One might suspect that the difficulty birds have recognizing transposed tone sequences reflects a general difficulty that animals have with recognizing sound sequences on the basis of *relations* between acoustic features (McDermott, 2009). However, such a view is challenged by the recent finding that at least one species of songbird (the European starling, *Sturnus vulgaris*) can readily learn to recognize frequency-shifted versions of songs from other starlings (Bregman, Patel, & Gentner, 2012). Such songs have complex patterns of timbre and rhythm, and the birds may recognize songs on the basis of timbral and rhythmic relations even when songs are shifted up or down in frequency. Yet when faced with isochronous tone sequences (which have no time-varying timbral or rhythmic patterns), the birds have great difficulty recognizing frequency-shifted versions. Hence they seem not to rely on relative pitch in tone sequence recognition, a striking difference from human auditory cognition.

Like birds, nonhuman mammals also do not seem to show a spontaneous reliance on relative pitch in tone sequence recognition. Some terrestrial mammals have been trained in the laboratory to recognize a single pitch interval (or even short melodies) shifted in absolute pitch (Wright, Rivera, Hulse, Shyan, & Neiworth, 2000; Yin, Fritz, & Shamma, 2010), but what is striking in these studies is the amount of training required to get even modest generalization, whereas human infants do this sort of generalization effortlessly and spontaneously (Plantinga & Trainor, 2005). Of course, many other species remain to be studied. Dolphins, for example, are excellent candidate for such studies, because they are highly intelligent social mammals that use learned tonal patterns in their vocalizations (McCowan & Reiss, 1997; Sayigh, Esch, Wells, & Janik, 2007; Tyack, 2008), and also have excellent frequency discrimination abilities (e.g., Thompson & Herman, 1975). A study of relative pitch perception in one bottlenose dolphin (*Tursiops truncatus*) showed that the animal could learn to discriminate short ascending from descending tone sequences after a good deal of training (Ralston & Herman, 1995). This work should be replicated and extended to see if

there are other cetacean species (other dolphin species, or belugas, orcas, etc.) that resemble humans in showing a spontaneous reliance on relative pitch in auditory sequence recognition. Such tests should employ species-specific sounds, such as dolphin signature whistles (Sayigh et al., 2007) as well as tone sequences (see Bregman et al., 2012 for this approach used with songbirds). If some cetaceans show a spontaneous reliance on relative pitch, and if nonhuman primates and birds don't show this trait, then this ability would be classified as "restricted to humans and select other species," and the finding would raise interesting questions related to convergent evolution (cf. the preceding section).

However, if this trait proves uniquely human, this would also raise interesting questions. Is the trait due to natural selection for musical behaviors in our species? Alternatively, might it be a consequence of the evolution of speech? In speech communication, different individuals can have very different average pitch ranges (e.g., men, women, and young children), and listeners must normalize across these differences in order to recognize similar intonation patterns spoken at different absolute pitch heights (such as a sentence-final rise, marking a question). Similarly, for speakers of tone languages to recognize the same lexical tones produced by men, women, and children, they must normalize across large differences in absolute pitch height to extract the common pitch contours and relations between pitches (Ladd, 2008; though cf. Deutsch, Henthorn, & Dolson, 2004 for a different view). Hence it is plausible that our facility with relative pitch is due to changes in human auditory processing driven by the evolution of speech.

Alternatively, our facility with relative pitch may be a developmental specialization of our auditory system, based on the need to exchange linguistic messages with conspecifics with a wide variety of pitch ranges. Perhaps we (like other animals) are born with a predisposition toward pitch sequence recognition based on absolute pitch cues, but this predisposition is overridden by early experience with our native communication system, that is, spoken language (Saffran, Reeck, Niebuhr, & Wilson, 2005). Were this the case, one might expect that all normal adult humans would retain some "residue" of absolute pitch ability, namely, an ability to recognize tone sequences on the basis of absolute pitch height. (Note that this type of absolute pitch is distinct from "musical absolute pitch," the rare ability to label isolated pitches with musical note names). In fact, recent studies show that normal human adults without musical absolute pitch simultaneously integrate relative and absolute pitch cues in music recognition (Creel & Tumlin, 2011; Schellenberg & Trehub, 2003; cf. Levitin, 1994). Interestingly, autistic individuals appear to give more weight to absolute pitch cues than normal individuals in both music and speech recognition, which may be one source of their communication problems in language (Heaton, 2009; Heaton, Davis, & Happe, 2008; Järvinen-Pasley, Pasley, & Heaton, 2008; Järvinen-Pasley, Wallace, Ramus, Happe, & Heaton, 2008). This fascinating issue clearly calls for further research.

How can one test the "speech specialization" theory against the "developmental experience" theory for our facility with relative pitch? One approach would be to continue to test other animals in relative pitch tasks (e.g., dolphins, dogs). If our facility with relative pitch is due to the evolution of speech, then no other animal

should show a spontaneous reliance on relative pitch in auditory sequence recognition, because speech is uniquely human. Another approach, however, is to attempt to provide other animals with early auditory experience that could bias them toward a reliance on relative pitch in recognizing sound patterns. For example, juvenile songbirds could be raised in an environment where pitch contour, as opposed to absolute pitch height, is behaviorally relevant (e.g., rising pitch contours indicate that a brief period of food access will be given soon, whereas falling contours indicate that no food is forthcoming, independent of the absolute pitch height of the contour). If this exposure is done early in the animal's life, before the sensitive period for auditory learning ends, might the animal spontaneously develop a facility for tone sequence recognition based on relative pitch? The idea that juvenile animals can develop complex sequencing abilities with greater facility than adults is supported by recent work with chimpanzees on visuomotor sequence tasks (Inoue & Matsuzawa, 2007; cf. Cook & Wilson, 2010). This idea leads to an important conceptual point for this section: before one can conclude that a component of music cognition is uniquely human, it is crucial to conduct developmental studies with other animals. Juvenile animals, who have heightened neural plasticity compared with adults, may be able to acquire abilities that their adult counterparts cannot. If an aspect of music cognition, such as a facility with relative pitch processing, cannot be acquired by juvenile animals, then this supports the idea that this aspect reflects evolutionary specializations of the human brain. Questions of domain-specificity then come to the fore, to determine whether the ability might have originated in another cognitive domain, such as language, or whether it may reflect an evolutionary specialization for music cognition.

E. Cross-Species Studies: Conclusion

About 25 years ago, Hulse and Page (1988) remarked that "research with animals on music perception has barely begun." The pace of research in this area has increased since that time, but the area is still a frontier within the larger discipline of music psychology. New findings and methods are beginning to emerge and are laying the foundation for much future research. This research is worth pursuing because cross-species studies can help illuminate the evolutionary and neurobiological foundations of our own musical abilities. Such research also helps us realize that aspects of music processing that we take for granted (e.g., our facility with relative pitch perception, or with synchronizing to a musical beat) are in fact quite rare capacities in the animal world, raising interesting questions about how and why our brains have these capacities.

III. Cross-Cultural Studies

A. Introduction

In cross-species comparative research, the groups under study (humans vs. other animals) often have very different cognitive capabilities, reflecting genetically

based differences in brain structure and function. By contrast, cross-cultural research begins with the assumption that all subject groups share the same intrinsic cognitive capabilities and that any differences in function must be due to the particularities of their experience. A neurologically normal infant born anywhere in the world could be adopted at birth and encultured into any existing musical culture without any special effort or training. This suggests that although there may be considerable surface differences in the musics of the world, they should share some fundamental organizational principles that relate to the predispositions and constraints of human cognition.

We find a similar situation in language. Humans have produced an astonishing array of linguistic systems that were developed using the same basic neural architecture. One key difference is that all known languages, even those that don't involve speaking, seem to share some universal grammatical characteristics (see Everett, 2005, for a possible exception). There has been no corresponding universal grammar of music proposed. This is not surprising when we consider that the communicative characteristics of music are far more ambiguous and polysemic than language (Slevc & Patel, 2011). This ambiguity permits a greater diversity of organizational possibilities than language. It also creates unique challenges in exploring potential similarities and differences in how music is made and perceived across different cultures. If we accept that all human cultures make music and that all neurologically normal humans share the same basic neural architecture, then what point is served by comparing the musical responses of subjects from different cultures?

Ethnomusicological research has at times been interested in the origins of music and in the possibility of universals in music. Unfortunately, the pursuit of comparative research into culture became entangled with notions of cultural evolution and the supposed superiority of some "developed" cultures (Nettl, 1983). Because of this association with ideas of cultural hegemony, ethnomusicology largely abandoned comparative research as inherently flawed, though some are beginning to reconsider the value of comparative work for clarifying cultural influences in musical thinking (Becker, 2004; Clayton, 2009; Nettl, 2000). There is general agreement that something with the general form and function of "music" exists in all known human cultures, so the very presence of music might be considered the first universal. After that starting point, however, things become much less clear. For example, ideas about what music *is* vary greatly from culture to culture so that even a cross-cultural definition of the word music is likely impossible (Cross, 2008). Nettl (2000) suggested that virtually all known musics have "A group of simple styles with limited scalar structure, and forms consisting of one or two repeated phrases" (p. 463). Nettl termed these features *statistical universals* because although they may not occur in absolutely every recognized culture, their presence is sufficiently ubiquitous to merit discussion (see Brown & Jordania, 2011 for an expansion of this idea). Clayton (2009) has argued that all of the world's musics may arise out of some combination of two characteristics, "vocal utterance and coordinated action" (p. 38). The challenge with identifying universal properties of music is that although we may inductively identify a large number of cultures

that feature such properties, deductively the absence of any property from even one musical tradition would call into question the notion of universality. Psychological approaches to exploring music universals, however, are not stymied by the lack of universal features of music across cultures, because they focus instead on the cognitive processes involved in musical thought and behavior. A number of authors have proposed processing universals that might function across cultures (Drake & Bertrand, 2001; Stevens & Byron, 2009; Trehub 2003). Processing universals derive from the shared cognitive systems used to perceive or produce music across cultures, even if the music produced by these shared processes sounds very different.

Cross-cultural music psychology offers a unique opportunity to test the validity of our thinking regarding fundamental processes of music cognition and their development through formal and informal means. Everybody has a unique biography of musical experiences. The degree to which informal musical experiences are shared by people growing up in a similar time and place constitute the construct of musical culture. Comparative research between cultures can provide a critical test of any theory that purports to explain human musical thinking in the broadest sense. If a theory of musical thought and behavior operates only within the constraints of one or even a few cultures, its utility as a universal explanatory framework is severely compromised. Two questions we can ask of any theory of music cognition are (1) Does it predict the behavior of listeners from any culture when encountering their own music? and (2) To what extent can it explain a listener's response to culturally unfamiliar music? The first question deals with universal *processes* in music cognition that might exist across cultures, whereas the second question points to *properties* of music that might transcend culture.

Comparative research also offers an opportunity to explore the distinction between innate and adaptable processes of music cognition. Infant research in particular has explored the possibility of innate predispositions for music processing (Trehub, 2000, 2003) and how those processes are shaped by culture in development. By exploring development cross-culturally, we can identify those aspects of music cognition that are differentiated by implicit learning of different musical systems and what aspects transcend cultural influences. A final purpose of comparative research in music cognition is to explore the influence of culture as a primary variable in music cognition. To what extent do cultural norms and preferences influence how the members of that culture perceive, produce, and respond to music?

Before reviewing the research in this field, it is useful to clarify what constitutes a "comparative" cross-cultural study in music psychology. The most basic kind of comparative study, what might be termed a partially comparative study, has participants from one culture (usually Western-born) respond to music of another culture, perhaps comparing those responses to responses on the same task using Western music. A variation of this partial design would be having participants from two cultures listening to the same music to compare their responses under the same condition. These studies, while useful, are incomplete because they do not establish the relevance of the variable under study or the judgment task for both cultures simultaneously. A fully comparative study includes both the music and the

participants of at least two distinct musical cultures. Such designs are less common in the field, but have yielded important results when they are employed because they help validate the relevance and representativeness of the variable under study in both cultures. These design distinctions should be kept in mind when evaluating the findings of cross-cultural research.

Although the body of research on the impact of culture on musical thinking is considerably smaller than in other areas of music psychology, its contributions to our understanding of music cognition and its development have been important. We will review several areas of comparative research that have contributed new perspectives to music psychology, including infant research, research on the perception of emotion, research on the perception of musical structure, and cognitive neuroscience approaches to exploring enculturation. Although a number of individual studies have employed cultural variables to some degree, the focus will be on programs of research that have explored cultural influences in multiple experiments.

B. Infant Research

One approach to exploring culture-general aspects of music cognition is to test the predispositions of infants for certain types of music processing. The assumption guiding this research is that infants are largely untouched by enculturation; therefore, any response preferences they exhibit might be assumed to be culturally neutral. Although this assumption can be questioned because auditory learning begins before birth (cf. Patel, 2008, pp. 377–387), it is reasonable to assume that infants are minimally encultured compared with adults. Hence infant predispositions for music might form the basis for identifying foundational processes of musical thinking that are eventually shaped by culture.

In two extensive reviews of infant research, Trehub (2000, 2003) proposed processes of music cognition that may be innate because infants seem predisposed to attend to those aspects of the musical stimulus. She observed that infants, like adults, can group tone sequences on the basis of similarities in pitch, loudness, and timbre; focus on relative pitch and timing cues for melodic processing; process scales of unequal step size more easily; show a preference for consonance over dissonance; and favor simpler versus more complex rhythmic information. It would seem that such predispositions might form a good starting point for examining cross-cultural similarities in music processing. By testing similar questions with infants and adults from several cultures, we might be able to form a better picture of how such predispositions interact with cultural experience and to what extent they can be altered by those experiences. For example, there may be a processing advantage for unequal scale steps, but this does not prevent the musical cultures of Java and Bali from developing equal-step scale systems. Would encultured members or even infants from those societies still exhibit the processing advantage for unequal scales?

One of the earliest examples of comparative infant research explored the role of culture and expertise in the perception of tuning by infants, children, and adults of varying experience (Lynch & Eilers, 1991, 1992; Lynch, Eilers, Oller, & Urbano,

1990; Lynch, Eilers, Oller, Urbano, & Wilson, 1991; Lynch, Short, & Chua, 1995). They asked listeners to identify when a deviant pitch (0.4%-2.8% change) appeared either on the fifth note of melodies based on major, minor, and *pelog* (Javanese pentatonic) scales or on a random note. Children and adults were better at detecting mistuned notes in culturally familiar stimuli (major and minor), though perceptual acuity differed by both age and training. In the first study, infants younger than 12 months were not influenced by cultural context, suggesting that their perceptual systems are open to a variety of input (Lynch et al., 1990); however, in later studies where the deviation position was variable, infants as young as 6 months performed better in a culturally familiar context (Lynch & Eilers, 1992; Lynch et al., 1995). The stimuli used in all of these studies were melodies based on extractions of original scale relationships using only notes 1 to 5 of the scale and presented in a uniform pure-tone timbre. A possibly more significant methodological issue was the decision to maintain the same absolute pitch level in the background melodies. Consequently, it is impossible to determine if infants were demonstrating sensitivity to deviations in relative or absolute pitch relationships. It would be useful to have this pioneering work replicated with some adjustments in both method and stimulus selection to critically test the findings.

Some of the most interesting comparative research being done with infants involves their sensitivity to cues associated with rhythmic and metrical grouping such as intensity and duration. Hannon and Trehub (2005a, 2005b) compared infant and adult ability to detect rhythmic changes to sequences set to isochronous (Western) and nonisochronous (Bulgarian) meters. In the first study (Hannon & Trehub, 2005a) they recorded the similarity ratings of Western and Bulgarian adults and Western infants to rhythmic variations in two metrical contexts (simple and complex) in three experiments. The variations either violated or preserved the original metrical structure. The simple meter featured 2:1 duration ratios typical of metrical structure in Western music and thought to be an innately preferred rhythmic bias in favor of simplicity (Povel & Essens, 1985). In Experiment 1, North American adults predictably rated the structure-violating variations as significantly more different, but only within the familiar metrical context. Their ratings of violations in the complex context did not differ on the basis of structural consistency. This result appears to confirm a processing bias for simple rhythms. However, in Experiment 2, Bulgarian and Macedonian-born adults rated the same stimuli. Because Bulgarian music frequently features irregular meters (e.g. $2 + 3$ or $3 + 2$ instead of $2 + 2$), this group responded identically to structure-violating variations in both metrical contexts, suggesting that cultural experience is more influential than a processing bias if one exists. In the third experiment, North American infants (6−7 months old) were tested on the same stimuli using a familiarization-preference paradigm that measured perceived novelty by recording looking time. The principle is that once habituated to a test stimulus, infants won't pay attention to the music source unless they hear a change. The degree of perceived novelty in that change is thought to correspond to the amount of time spent looking at the sound source. The infants were sensitive to structure violating variations in both metrical contexts disproving the hypothesis of any intrinsic processing bias

for simple meters. In addition to disproving a perceptual bias hypothesis, the research provided support for the assumption that infants less than 1 year old do not demonstrate a cultural bias in their processing as their performance was more similar to the Macedonian adult group than the North American adult group. A subsequent study (Hannon & Trehub, 2005b), tested responses of 11- to 12-month-old infants in two experiments. In Experiment 1, older infants demonstrated a cultural bias in their responses similar to the North American adults of the previous study. In the second experiment, infants were again tested but after brief at-home exposure (15-minute CD twice a day) to the irregular meters of Balkan dance music. The infants exposed to Balkan music did not demonstrate the same cultural bias for Western music as their uninitiated counterparts suggesting that brief exposure at this age can reverse the cultural bias of enculturation. Such exposure did not significantly reverse the cultural bias of adult participants who completed 2 weeks of a similar listening exposure in a pre-post design in Experiment 3. These two studies, simultaneously employing a culture-based and age-based comparison, elegantly parsed the relative influence of innate, encultured, and deliberate experience. In a subsequent study (Soley & Hannon, 2010), North American and Turkish infants age 4–8 months were tested for their preference for music employing Western or Balkan meters. The monocultural Western infants preferred Western metrical examples even at this young age, whereas the Turkish infants, who likely were exposed to both types of music, showed no preference. Both groups preferred real metrical examples to examples in an artificial complex meter, suggesting a possible bias for simplicity found in another study (Hannon, Soley & Levine, 2011). These studies provide a nice model for future investigations of this type because they offer fully comparative designs and feature the rare inclusion of non-Western infants (see also Yoshida, Iversen, Patel, Mazuka, Nito, Gervain, & Werker, 2010).

As Gestalt psychologists observed, human beings are expert pattern detectors. Although infants start with the same species-specific cognitive resources and predispositions for language and music, their performance appears to be influenced by the implicit learning of cultural norms at a very early age. Findings indicate that infants retain some flexibility even after demonstrating a cultural bias, whereas adults appear incapable of a similar flexibility. Although the concept of enculturation is widely accepted, the process by which it occurs is not well understood. Research in language development by Saffran and colleagues (McMullen & Saffran, 2004; Saffran, Aslin, & Newport, 1996) has identified a process of statistical learning that may explain how different cultural systems of music and language are learned implicitly. Although transitional probabilities have been manipulated in artificial music stimuli (Saffran, Johnson, Aslin, & Newport, 1999), it would be interesting to see if differences in transitional probabilities in extant melodies from different cultures could be quantified and used to predict cross-cultural responses to music or to track the process of enculturation in infant development as has been done with language (Pelucchi, Hay, & Saffran, 2009).

Comparative research with infants, especially with infants from multiple cultures, has tremendous potential for clarifying how culture impacts cognitive development by identifying both shared processes and points of differentiation.

We know that individuals can be bimusical just as they are bilingual, but are there similar critical periods for musical category development, or is music more fluid between cultures than language? The techniques of cognitive neuroscience, particularly electroencephalography/magnetoencephalography measurements, are being used increasingly in infant research to measure responses to music at very young ages (Winkler, Haden, Ladinig, Sziller & Honing, 2009). These techniques may allow us to compare infants' responses earlier and more reliably as they encounter culturally unfamiliar stimuli at various stages of development.

C. Perception of Emotion

One of the challenges inherent in cross-cultural research in music is the lack of clear meanings ascribed to musical utterances. The ambiguity of any semantic content in the musical utterance no doubt accounts for the popular belief in music as a universal language. After all, who can say that one's culturally naïve interpretation of music is wrong? Research into the perception of emotion in music has posited predictable shared meanings for musical utterances within a culture. There is considerable evidence that acoustic cues like tempo, loudness, and complexity can influence basic emotional judgments (joy/sadness) of music (Dalla Bella, Peretz, Rousseau, & Gosselin, 2001; Juslin, 2000, 2001; Juslin & Laukka, 2000, 2003). These acoustic properties are not solely musical but may mimic physical aspects of emotional behavior and prosodic expressions of emotion in language. To the extent that these properties are domain-general, musical representations of emotions may transcend culture by tapping into more fundamental responses to the human condition.

Balkwill and Thompson (1999) proposed a cue-redundancy model (CRM) of emotion recognition in music based on information from two kinds of cues: psychophysical cues were defined as "any property of sound that can be perceived independent of musical experience, knowledge or enculturation" (p. 44). Properties like rhythmic or melodic complexity, intensity, tempo, and contour are examples of psychophysical cues. For cultural outsiders, it was these cues alone that would allow them to recognize emotional representations in music outside of their culture. For a cultural insider, they proposed that these cues interacted redundantly with a second set of culture-specific cues such as instrumentation or idiomatic melodic/harmonic devices that reinforce the emotional representation. Cue redundancy (Figure 2) could account for outsiders' ability to perceive emotional content across cultures while retaining insider advantage for music of their own culture. The authors have more recently proposed a fractionating emotional systems model to describe a process of cross-cultural emotion recognition in both music and speech prosody as well as how those two systems might interact (Thompson & Balkwill, 2010).

Research in the area of cross-cultural perceptions of emotion in music has explored the affective judgments of both adults (Balkwill, 2006; Balkwill & Thompson, 1999; Balkwill, Thompson & Matsunaga, 2004; Deva & Vermani, 1975; Fritz et al., 2009; Gregory & Varney, 1996; Keil & Keil, 1966) and children

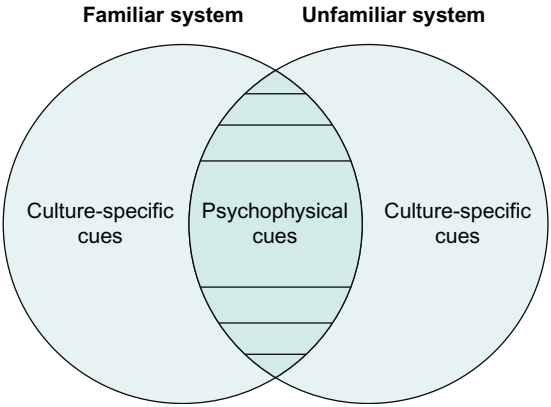

Figure 2 The cue-redundancy model (CRM) proposed by Balkwill and Thompson (1999). See text for details.
Reproduced with permission from Thompson and Balkwill (2010).

(Adachi, Trehub, & Abe, 2004). Comparative research was an early interest of ethnomusicologists, and one of the earliest studies to explore the cross-cultural perception of emotional meaning was published in an ethnomusicology journal (Keil & Keil, 1966). This study, along with Deva and Virmani (1975), used semantic differential methods to explore Western and Indian listeners' responses to Indian ragas to see if theoretical claims about intended emotion could be confirmed by listener judgments. Although there was agreement on certain melodies, there was great variability on others both within and between cultures. Gregory and Varney (1996) directly compared the responses of listeners from Western (British) and Indian heritage to Western classical music, Western new age music, and Hindistani ragas. They used the Hevner adjective scale to see if listeners could identify the emotions intended by the composers of the pieces. They reported general agreement in adjective choice between Western and Indian listeners on Western music, but not on Indian music, and they concluded that subjects could not accurately determine the mood intended by the composer. Their results are complicated by several factors: (1) their sample compared monocultural Western listeners to bicultural Indian listeners, (2) there was not an equal number of examples from each culture, and (3) the intended mood of the pieces was not determined through listener judgment but was "inferred by the authors from the title of the piece, descriptions of the music by writers or musicians and, for the Indian ragas, from the descriptions given by Danièlou" (pp. 48–49). All of these factors make it difficult to determine to what extent culture played a role in the judgments of the listeners because in-culture agreement seemed problematic as well.

Balkwill and Thompson (1999) had 30 Canadian listeners rate the emotional content of 12 Hindustani ragas that were theoretically associated with the four emotions of joy, sadness, anger, and peace. The listeners heard the ragas in a random order, were asked to choose one of the four emotions in a forced-choice model, and then rate on a scale from 1 to 9 the extent to which they felt that

emotion was communicated. They were able to clearly identify the ragas associated with joy and sadness, and their ratings correlated significantly with the ratings of four cultural experts. The data for anger and peace were less distinct both within the outsider group and between experts and novices. As the cue redundancy model suggested, ratings were associated with psychophysical properties. Joy ratings correlated with low melodic complexity and high tempo, whereas sadness ratings were based on the opposite combination.

Two subsequent studies expanded on the first by having Japanese listeners (Balkwill et al., 2004) and Canadian listeners (Balkwill, 2006) rate the emotional content of Japanese, Western, and Hindustani music. This time the choices were reduced to three emotions: anger, joy, and sadness. They found agreement across the three music cultures for all three emotions on the basis of psychophysical properties, but the Canadian listeners did differ from the Japanese in the cues associated with anger. The Japanese listeners used a broader combination of cues to make their judgments, which the authors suggest may reflect a cultural preference for more holistic processing identified in other research. It is interesting to note that the studies of emotion recognition that feature better agreement between (and within) cultures are those that limit responses to only a few broad categories rather than more sensitive descriptive measures. This may reflect the limitations of music's denotative power or may reflect a broader constraint of two-dimensional theories of emotion.

In these studies, Hindustani music provided the cultural "other" because it was a well-developed but less disseminated music culture than Western art music. A number of authors (Demorest & Morrison, 2003; Thompson & Balkwill, 2010) have cautioned against the use of Western music as an unfamiliar stimulus for any group given its ubiquity in commercial music across the globe. Fritz and colleagues (2009) explored emotion recognition responses to Western music with a sample of 20 German listeners and 21 members of the culturally isolated Mafa tribe in Northern Cameroon. Because of the Mafa's geographic isolation and lack of electrical power, the authors were confident that they were unfamiliar with Western music. They used short piano pieces chosen to represent one of three emotions (happy, sad, scared/fearful). All participants responded by choosing one of the three emotions from a nonverbal pictorial task featuring the facial expressions of a white female. Both groups were able to identify the intended emotions at better than chance level, though the variability in the Mafa subjects was much greater (including two subjects who did perform at chance level). There were no corresponding examples of Mafa music to compare cultural tendencies in that direction. Rating tendencies suggested that both groups used temporal and mode cues to make their judgments, though the tendency was stronger with in-culture listeners. They suggest that both groups may be relying on acoustic cues in Western music that mimic similar emotion-specific cues in speech prosody.

The connection of emotional communication in music to the characteristics of emotional speech has been posited by a number of researchers and suggests that any mechanism for identifying emotional representations in music may not be domain specific (cf. Juslin & Laukka, 2003). Like recognition of frequency of occurrence and transitional probability of notes in tonality, emotion recognition may rely on

general perceptual mechanisms that operate across domains. If so, then a unified theory of emotion recognition across musical, linguistic, and possibly even visual domains should be possible and might go further in explaining how humans across cultures express shared physical and emotional states through different modalities.

D. Perception of Musical Structure

Numerous writers have suggested that there are aspects of musical structure and cognition that are universal across cultures. Although some have focused on the features shared by many of the world's musics (Brown & Jordania, 2011; Nettl, 2000), others have focused on possible universal processes of music cognition (Drake & Bertrand, 2001; Stevens & Byron, 2009). Some of the candidates for processing universals are those evident in general cognition such as grouping events by the Gestalt principles of proximity, similarity, and common fate. Stevens and Byron (2009) suggest a list of possible universals in pitch and rhythm processing that "await further cross-cultural scrutiny," including pitch extraction, discrete pitch levels, the semitone as the smallest scale interval, unequal scale steps, predisposition for small integer frequency ratios (2:1, 4:3), octave equivalence; memory limitations in rhythmic grouping, synchronizing to a beat; and small integer durations (p. 16). Many of these possible "universals" were originally proposed from results of research with culturally narrow samples, but are beginning to be explored in both cross-cultural and cross-species research. This section presents some comparative studies that deal with the perception of pitch structure in melodies.

Comprehending higher-level melodic structure depends on perceiving fundamental relationships, but also requires listeners to retain numerous pitch and rhythm events in memory and to continually group and organize them over time as they listen. The perception of larger structural relationships also involves prediction of what comes next, i.e., a listener's musical expectations (Huron, 2006; Meyer, 1956; Narmour, 1990, 1992). These expectations are formed and refined through exposure to music and thus are likely to be more dependent on prior cultural experience than the more fundamental aspects of pitch and rhythm processing. Huron (2006) identifies three types of expectations, schematic, veridical, and dynamic. Schematic expectations are not specific to a certain piece or pieces, but are top-down general "rules" for music developed through exposure to a broad variety of music within a culture or cultures. Veridical expectancies are those associated with knowledge of a particular piece of music or musical material. Dynamic expectancies are the most bottom-up expectations, reflecting the moment-to-moment expectations formed while listening to a piece of music. The interaction between schematic and dynamic expectation determines our responses to newly encountered music of various styles and genres. Researchers have explored the perception of musical structures cross-culturally in a variety of ways.

One of the central aspects of melodic structure in pitch-based systems is the concept of tonality, or the grouping of pitches within a scale hierarchically. Tonal hierarchy theory (Krumhansl & Kessler, 1982; Krumhansl & Shepard, 1979) seeks to explain the music theoretic construct of tonality from a perceptual standpoint.

To test this theory in Western music, Krumhansl and Shepard (1979) developed the probe tone method. Listeners first hear tones that create a musical context, such as a major scale, melody, tonic chord, or chord sequence. After hearing this context, subjects then hear a single pitch or "probe" stimulus. Subjects are asked to rate how well they thought the probe tone fit into or completed the prior musical context. Tonal hierarchy theory has predicted Western listeners' responses to tonal relationships in a variety of contexts, but has also been tested in non-Western contexts.

Castellano, Bharucha, and Krumhansl (1984) tested the predictions of tonal hierarchy theory using the music of north India. North Indian music was chosen because it has a strong theoretical tradition that posits relationships between tones, but those relationships develop melodically rather than harmonically. The researchers tested both Western and Indian listeners responses to 10 North Indian rãgs and found that both groups were sensitive to the anchoring tones of the tonic and fifth scale degrees and gave stronger stability ratings to the vãdi tone, the tone given emphasis in each individual rãg. Only the Indian listeners, however, were sensitive to the thãts or scales underlying each rãg, suggesting that prior cultural experience was necessary to recover the underlying scale structure of the music.

Kessler, Hansen, and Shepard (1984) used stimuli and subjects from Indonesia and the United States. They compared responses of all subject groups to Western major and minor musical scales and two types of Balinese scales (pelog and slendro). They found that subjects used culturally based schema in response to music of their own culture, but used a more global response strategy when approaching culturally unfamiliar music that concentrated on cues such as frequency of occurrence for a particular tone.

Even though there was some advantage for those with insider cultural knowledge, Krumhansl summarized the findings for the two studies by concluding, "In no case was there evidence of residual influences of the style more familiar to the listeners on ratings of how well the probe tones fit with the musical contexts" (1990, p. 268). Since that time, there have been subsequent cross-cultural studies with Chinese music (Krumhansl, 1995), Finnish folk hymns (Krumhansl, Louhivuori, Toiviainen, Jarvinen, & Eerola, 1999), and Sami yoiks (Krumhansl et al., 2000) that have yielded more mixed results with regard to the cultural transcendence of tonal perception. The findings from the more recent research suggest that the perception of tonality involves a combination of bottom-up responses to the stimulus involving the frequency of occurrence for tones or their proximity in a melody, as well as top-down responses that are informed by subjects' prior cultural knowledge. In the cases where subjects' cultural schema do not fit, their judgments can mimic an insider's up to a point, and then they diverge. For example, in the studies using longer examples of Finnish and Sami melodies, Western listeners were able to make continuation judgments that reflected the general distribution of tones heard up to that point, but were not able to completely suppress their style-inappropriate expectancies and differed significantly in certain judgments from those subjects who were experts in the style (Krumhansl et al., 1999, 2000).

In the studies cited previously, the authors were interested primarily in whether outsiders could detect tonal hierarchies in culturally unfamiliar music. In a more

recent study, Curtis and Bharucha (2009) sought to exploit culturally based schemata to fool Western-born listeners into an incorrect judgment. They used a recognition memory paradigm similar to those used in false memory research. They presented listeners with one of two tonal sets based on a Western major mode (Do Re Mi Fa Sol La Ti) or the Indian thãt Bhairav (Do Re- Mi Fa Sol La- Ti), which shares all but two notes with the other scale. Each scale was presented as a melody missing either the second or sixth scale degree (e.g., Fa Mi Do Re- Sol Ti Do for Bhairav). Each presentation was followed by a test tone that was either the tone that was present in the tone set (Re- in Bhairav), the missing tone that was musically related (e.g., La- in Bhairav), or the tone that was musically unrelated to the tone set (e.g., La or Re in Bhairav). The prediction was that listeners would incorrectly "remember" the musically related tone that was missing, but only in the culture with which they were familiar. In trials where the test tone had occurred (25%), subjects were equally accurate at recognizing that they had heard the tone regardless of culture. In trials where the test tone had not occurred (75%), Western modal knowledge biased subjects' responses so that they falsely "remembered" hearing the tone from the Western set (Re/La). This was particularly true when a Western test tone was played for an Indian scale set, suggesting that cultural learning plays a role in the melodic expectancies we generate. This cultural bias has also been demonstrated neurologically in studies of expectancy presented later in the chapter.

Although infant research has begun to explore the role of culture in rhythmic development, there are relatively few studies of adult rhythmic processing from a cross-cultural perspective. Individual studies have explored the influence of enculturation in synchronization (Drake & Ben El Heni, 2003), cultural influences on the meter perception and the production of downbeats (Stobart & Cross, 2000), and melodic complexity judgments (Eerola, Himberg, Toiviainen, & Louhivuori, 2006). Several studies have explored the relationships between the musical and linguistic rhythms in a culture. Patel and Daniele (2003) applied a quantitative measure developed for speech rhythm to analyze durational patterns in the instrumental music of French and British composers. They found a relationship between the musical rhythms and the language of the composer's origin. Subsequent research has established that musical rhythms can be classified by language of origin (Hannon, 2009) and that linguistic background can influence the rhythmic grouping of nonlinguistic tones in adults (Iversen, Patel, & Ohgushi, 2008) and infants (Yoshida, et al., 2010) from different cultures.

E. Culture and Musical Memory

If we want to identify where musical understanding breaks down between cultures, then how does one measure the "understanding" of music? One idea is to study musical memory. Musical memory requires one to group or chunk incoming information into meaningful units, and this process is influenced by prior experience (e.g., Ayari & McAdams, 2003; Yoshida et al., 2010). Several studies have explored the impact of enculturation on broader musical understanding as

represented by memory performance (Demorest, Morrison, Beken, & Jungbluth, 2008; Demorest, Morrison, Stambaugh, Beken, Richards, & Johnson, 2010; Morrison, Demorest, Aylward, Cramer, & Maravilla, 2003; Morrison, Demorest, & Stambaugh, 2008; Wong, Roy & Margulis, 2009). In all of these studies, recognition memory was used as a dependent measure of subjects' ability to process and retain the different music styles they were hearing. Memory was chosen because (1) it is not culturally biased, (2) it allows the use of more ecologically valid stimuli, and (3) better memory performance can indicate greater familiarity or understanding. The hypothesis was that if schemata for music are culturally derived, then listeners should demonstrate better memory performance for novel music from their own culture than that of other cultures.

One fully comparative study (Demorest et al., 2008) tested the cross-cultural musical understanding of musically trained and untrained adults from the United States and Turkey. Participants listened to novel music examples from Western (U.S. home culture), Turkish (Turkish home culture), and Chinese music (unfamiliar control) traditions. Memory performance of both trained and untrained listeners was significantly better for their native culture, a finding they dubbed the "enculturation" effect. Turkish participants were also significantly better at remembering Western music than Chinese music, suggesting a secondary enculturation effect for Western music. In all conditions, formal training in music had no significant effect on memory performance. A subsequent study compared the memory performance of U.S.-born adults and fifth-graders listening to Western and Turkish music and found a similar enculturation effect for their home music across two levels of musical complexity with no significant differences in performance by age (Morrison et al., 2008). The generalizing of this effect to younger subjects and to music of varying complexity suggests that enculturation has a powerful influence on our schema for music structure.

Wong et al. (2009) compared the responses of three groups; monocultural U.S. listeners, monocultural Indian listeners and bicultural Indian listeners on two cross-cultural tasks. The first task was a recognition memory task similar to those used in previous studies, but using Western and north Indian melodies. The second task was a measure of perceived tension in Western and Indian music. In both tasks monocultural subjects demonstrated a positive performance bias (better memory, lower perceived tension) for music of their own culture, with the bimusical individuals showing no differentiation on either task. This is one of the first studies to test the concept of bimusicality empirically in a controlled study. Memory structures seem to be powerfully influenced by prior cultural experience. Future research might explore how easily such structures are altered by short-term exposure and what types of experiences might influence or equate memory performance between cultures.

F. Cognitive Neuroscience Approaches

The research presented thus far has relied on measuring subjects' behavioral responses to music under different conditions. As mentioned earlier, such conscious

responses to musical information are a challenge for cross-cultural research, where the task itself may be biased toward one culture's world view. Neuroscience approaches to comparative research offer researchers another window on cognition that can complement the information they are receiving from subjects' behavior. Comparative studies employing neuroscience approaches have explored a number of topics already mentioned, including the cross-cultural perception of scale structure (Neuhaus, 2003; Renninger, Wilson, & Donchin, 2006), phrase boundaries (Nan, Knösche, & Friederici, 2006; Nan, Knösche, Zysset, & Friederici, 2008), tone perception related to native language (Klein, Zatorre, Milner, & Zhao, 2001), culture-specific responses to instrument timbre (Arikan, Devrim, Oran, Inan, Elhih, & Demiralp, 1999; Genç, Genç, Tastekin, & Iihan, 2001), cross-cultural memory performance (Demorest et al., 2010; Morrison et al., 2003), and bimusicalism (Wong, Chan, Roy, & Margulis, 2011).

Comparative studies of tonal hierarchy mentioned earlier indicated that listeners exhibited hierarchical responses to culturally unfamiliar music, but only in response to the distribution of tones heard previously in the context. Cultural background was revealed when subjects made judgments that required an understanding of the background tonality induced by the context (Castellano et al., 1984; Curtis & Bharucha, 2009; Krumhansl et al., 1999, 2000). Cross-cultural sensitivity to tonality violations has been explored by examining Event-Related Potential (ERP) responses to scale violations in familiar and unfamiliar scale contexts using an oddball paradigm where scale notes were presented continuously with nonscale notes interspersed as oddballs (Neuhaus, 2003; Renninger et al., 2006). In both studies, they found that listeners were not sensitive to tonality violations for unfamiliar cultures unless such a violation conformed to their culture-specific expectancies.

The ERP method has tremendous potential for illuminating culture-specific differences in expectancy and offers the opportunity to test both bottom-up and top-down of models of expectancy formation against subjects' neurological responses to violations. It will be important for future research to compare intact melodies rather than isolated scales. Ultimately it would be desirable to develop theoretical models of expectancy in different cultures, a measure of the cultural "distance" between two systems that could be used to predict listeners' responses on the basis of their cultural background. Developing databases of non-Western melodies similar to the Essen Folksong Collection for Western music (Schaffrath, 1995) may provide the raw material for charting differences in transitional probabilities of pitch content or rhythmic patterns between cultures. ERP might also be used to explore cross-cultural music learning using methods similar to those for exploring second language learning (McLaughlin, Osterhout, & Kim, 2004).

As mentioned before, memory is another area thought to rely heavily on culturally derived schemata for music. The influence of enculturation on music memory has been explored in two functional magnetic resonance imaging (fMRI) studies (Demorest et al., 2010; Morrison et al., 2003). In the first study, Western-born subjects, both musically trained and untrained, were presented with three 30-second excerpts from Western art music interspersed with three excerpts from Chinese traditional music and then three excerpts of English-language and Cantonese language

news broadcasts. The hypothesis was that there would be significant differences in brain activity for culturally familiar and unfamiliar music and language based on differences in comprehension. They found a difference for linguistic stimuli but not musical stimuli, though there were significant differences in expert/novice brain responses and differences by musical culture in a memory test that subjects took after leaving the scanner. To explore the discrepancy between the behavioral and neurological findings of the first study, Demorest et al. (2010) had U.S. and Turkish born subjects listen to excerpts from three cultures, Western art music, Turkish art music, and Chinese traditional music. After each group of stimuli, subjects took a 12-item memory test in the scanner. Brain activity for both subject groups was analyzed by comparing responses to their home music (Western or Turkish, respectively) with a musical culture unfamiliar to both (Chinese). They found significant differences in brain activation in both the listening and the memory portion of the task based on cultural familiarity. Although both tasks activated the same network of frontal and parietal regions, the activation was significantly greater for the culturally unfamiliar music. The authors interpreted this increase in activation as representing a greater cognitive load when processing music that does not conform to preexisting schemata. Nan et al. (2008) found a similar difference in activation when subjects engaged in a phrase-processing task in an unfamiliar culture.

Phrase processing was also explored in a fully comparative ERP study (Nan et al., 2006) with highly trained German and Chinese musicians. Researchers were investigating whether out-of-culture listeners would exhibit a closure positive shift (CPS) that occurs between 450 and 600 milliseconds after an event and has been used to measure sensitivity to boundaries in both music and language. Stimuli for the study were little-known eight-bar phrases taken from Chinese and German melodies and presented in a synthesized piano timbre and in either phrased or unphrased version for each culture. Behaviorally both groups exhibited superior performance within their native style. Despite differences in behavioral performance, all subjects demonstrated a CPS response to phrased melodies from both cultures, similar to findings for within-culture studies (Knösche et al., 2005; Neuhaus et al., 2006). German subjects did exhibit larger responses to Chinese music deviants at earlier latencies, suggesting some conflict between task demands and enculturation. There was no corresponding difference for the Chinese musicians who were familiar with Western music.

Building on an earlier behavioral study of bimusicalism, Wong and colleagues (Wong et al., 2011) scanned bimusical (Western and Indian) and monomusical (Western only) subjects while they made continuous tension judgments for Western and Indian melodies. They used structural equation modeling (SEM) to examine connectivity among brain regions and correlations to the behavioral measure. The results suggest that monomusicals and bimusicals process affective musical judgments in qualitatively different ways. The application of neuroimaging techniques to questions of culture is a relatively new but growing field (Chiao & Ambady, 2007; Morrison & Demorest, 2009), one that holds great promise for unlocking the complex interplay of perception and cultural experience.

G. Cross-Cultural Studies: Conclusion and Considerations for Future Research

The role of cultural experience in music perception and cognition is complex, involving an interplay of bottom-up, global perceptual mechanisms that respond to the distribution of tones, durations, and contours of a musical stimulus with top-down culturally learned schemata that guide how such information is combined into meaningful units. The promise of comparative cross-cultural research is that it can help tease out the relative influence of those competing systems to provide a more complete picture of the mechanisms of music perception. It may also hold the key to uncovering domain-general perceptual processes that operate across cultures and across modalities such as music, language, and vision. Almost any theory or research question that has been explored within a Western cultural framework might be reexamined from a comparative perspective. Future research needs to be conscious of the methodological challenges of cross-cultural comparative research and begin to connect the work in music to strong theoretical models of cultural influence within and between disciplines.

There are a few methodological considerations that can help researchers avoid common pitfalls of cross-cultural research. First, both the tasks and the stimuli used in a comparative study should be legitimate in both cultures. One way to ensure this is to include members of all cultures under study in the subject pool (fully comparative studies) and on the research team that puts the design together. A second concern is the role of context. Ecological validity has long been a concern in empirical research, but the relative importance of musical context can differ by culture. For example, in some cultures it would be unusual to listen to music without an accompanying dance or movement of some kind. Consequently, the implications of removing contextual variables for experimental control in a comparative study may differentially influence subject responses, thereby skewing results. Context and its potential manipulation needs to be a consideration in any culturally comparative study of music cognition.

Successful applications of theoretical models and techniques from language and emotion research suggest that at least some mechanisms of music perception are not domain specific (Patel, 2008; Saffran, Johnson, Aslin, & Newport, 1999; Thompson & Balkwill, 2010). Merker (2006) concluded "a cautious interpretation of the evidence regarding human music perception contains few robust indications that humans are equipped with species-specific perceptual-cognitive specializations dedicated to musical stimuli specifically. That is, the evidence reviewed does not force us to conclude that selection pressures for music perception played a significant role in our evolutionary past." (p. 95). Researchers interested in cross-cultural music cognition research might look to comparative research in other domains for possible domain-general models of culturally influenced cognitive processing. Research in this area would also benefit from stronger musical models such as information-theoretic analyses of musical content that might predict listener responses or theories of music-motor connections that might be affected by cultural connections between music and movement. Equally important is that researchers

focus on opportunities to disprove rather than confirm theories of universality in music cognition by carefully selecting comparisons that, on the surface, should yield differences by culture. For example, the notion of a preference for simple (2:1) ratios in meter was conclusively disproven by a comparative study, whereas emotion recognition seems to rely on some culturally transcendent features. Many other proposed universals (Brown & Jordania, 2011; Drake & Bertrand, 2001; Nettl, 2000; Stevens & Byron, 2009; Trehub, 2003) await comparative testing.

IV. Conclusion

It has been roughly three decades since the first edition of *The Psychology of Music*, and more than a decade since the foundational chapter by Carterette and Kendall (1999) on comparative music perception and cognition in the second edition. During that time, research that looks beyond our own species and beyond Western culture has grown considerably. Nevertheless, these are still frontier areas within music psychology, with relatively small bodies of research when compared with the literature on human processing of Western tonal music. In this chapter, we have argued that comparative studies of music cognition are essential for studying the evolutionary history of our musical abilities, and for studying how culture shapes our basic musical capacities into the diverse forms that music takes across human societies. From the standpoint of psychology, the fact that certain aspects of music do cross species and cultural lines, while others do not, makes comparative music cognition a fascinating area for studying how our minds work. Humans are biological organisms with rich symbolic and cultural capacities. A full understanding of music cognition must unify the study of biology and culture, and in pursuing this goal, comparative studies have a central role to play.

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